

Environmental Kuznets curves: A spatial econometric approach

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Abstract

Most researchers estimating environmental Kuznets curves (EKC) implicitly assume that countries' per capita emissions are unaffected by events in neighbouring states. Using a panel of sulphur dioxide, nitrogen oxides, volatile organic compounds and carbon monoxide emissions, a conventional EKC is augmented by spatially weighted values of the dependent and independent variables. National emissions per capita of sulphur dioxide and nitrogen oxides are shown to be heavily influenced by the emissions per capita of neighbouring countries. There is also evidence that national emissions per capita of nitrogen oxides are decreased by proximity to high per capita income countries which is inconsistent with nations achieving higher environmental quality at the expense of their neighbours.

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1. Introduction

Interest in environmental Kuznets curves (EKC) shows little sign of diminishing. This is unsurprising given the overriding importance of the issue that this literature seeks to address. One view is that economic growth will necessarily require more energy and materials leading to ever greater levels of environmental degradation. An alternative view suggests that economic growth may improve environmental quality through technological change and economies of scale in pollution abatement, alterations in the composition of output and an increasing demand for environmental quality. If this latter view is correct then one might expect emissions initially to grow when a poor country begins to expand its production. Eventually, however, a turning point would be reached after which pollution per capita would decrease. The question at stake is whether economic growth is ultimately compatible with environmental quality.

EKCs have now been estimated for a variety of pollutants including sulphur dioxide, carbon dioxide, biological oxygen demand, nitrogen oxides, smoke and particulate matter. At its most basic the technique involves regressing per capita emissions or concentrations on income per capita, its squared value and a time trend. Early empirical studies appeared to confirm the view that environmental quality rises with respect to per capita income and then declines consistent with the more optimistic outlook, although there are now as many papers obtaining different outcomes.

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Given the existence of numerous reviews of the EKC literature [14,15,17,21,43,45] this paper does not attempt to go over the empirical evidence again. To the casual observer it nevertheless appears that researchers have recently started to pay much closer attention to the temporal properties of the data examining issues such as non-stationarity, co-integration, autocorrelation and partial adjustment [18,27,35,40]. Stationarity in particular is an important feature of any data set since, without this characteristic, a regression of pollution on per capita income risks generating spurious results.

But contrast the special treatment afforded to time series data with that given to spatial data. In reality, most data sets used to estimate EKCs contain elements of both being composed of repeated observations on *countries*. It does not seem implausible to suggest that observations drawn from geographically neighbouring locations might be more closely related than those taken from geographically distant ones. But unlike with tests for stationarity, etc., checks for proximity-based relationships are seldom if ever performed in the EKC literature. The existence of spatial relationships in the data nevertheless has important implications for the econometric techniques typically employed including faulty inference testing procedures and possible bias [8].

In EKCs, spatial relationships arise most obviously as a consequence of countries' strategic response to transboundary pollution flows. Two papers [38,44] have attempted empirically to account for spatial relationships between countries arising from the transboundary deposition of sulphur in Europe although only the latter explicitly addresses the EKC.¹ But there are in fact numerous other reasons why spatial relationships might be present in the data used to estimate EKCs. Several authors have suggested, e.g., that the shape of the EKC is a consequence of high-income countries in effect exporting their pollution to lower-income countries [41,47]. If the 'pollution displacement' hypothesis is correct a poor country immediately neighbouring a rich one would more likely have higher emissions given that numerous empirical analyses [1] have identified distance and the existence of common land borders as important factors in facilitating trade. Recently reviewed empirical evidence [30] suggests that geography is likewise a major determinant of technology diffusion. To the extent that improvements in per capita emissions occur as a result of technological change one might expect to find spatial patterns in per capita emissions.²

A large body of literature suggests that governments might strategically manipulate environmental standards in an attempt to attract capital, or for trade purposes [37,48]. The pursuit of either implies that changes in environmental policies might result from changes in neighbouring countries. Government officials and bureaucrats are also constantly assessing policy against those of their neighbours in order to reduce the costs of decision-making and to legitimise their actions, particularly when there is uncertainty regarding the effects of those policies. This might result in countries mimicking each other environmental policies resulting in similar environmental standards. It has been shown that states in the US raise their environmental standards in response to improvements in the environmental standards achieved by neighbouring states [25].³ The idea that environmental policies promulgated in one country spill over into other countries also finds support in the sociological literature [24].

Although it may in practice prove difficult to pin down the precise reason for spatial relationships in the data their existence would nevertheless provide a potential explanation of why many authors have observed instability in the parameters of the EKC. One of the main implications of spatial spillovers is that countries behave differently depending on the characteristics of their neighbours. Incorrectly omitting spatially lagged variables may cause the parameters of the EKC to become biased. Stern and Common [46] find that when they use a global sample of countries there is a monotonically increasing relationship between emissions per capita and per capita income. But when the data set is restricted to high-income countries they obtain an inverse U-shaped relationship. In their study of carbon dioxide emissions, Dijkgraaf and Vollebergh [20] analyse emissions for a panel of countries and determine that there exists an inverted U-shaped relationship. However, when EKCs are separately estimated for each of the countries a very different set of results emerges prompting the authors to question the existence of a global EKC.

¹See also [9].

²Most low- and middle-income countries acquire more environmentally friendly technology from abroad [34].

³Interestingly, the relationship is shown to be asymmetric in that a state is more likely to raise its environmental standards in a response to a situation in which neighbours have higher standards than to reduce them in response to a situation in which neighbours have lower standards.

Somewhat surprisingly contemporary reviews of the methodological and econometric issues involved in the estimation of EKC's do not mention spatial econometrics as a currently unaddressed research issue.⁴ Only two papers have estimated EKC's using spatial econometric techniques. But even the most recent of these models the spatial relationships in the data as a “nuisance” rather than a “substantive” process.⁵

The purpose of this paper is to explore systematically the use of spatial econometric techniques in estimating EKC's. A conventional EKC is estimated for national per capita emissions of sulphur dioxide, nitrogen oxides, non-methane volatile organic compounds (VOCs) and carbon monoxide. Using repeated observations, EKC's are estimated in first differences to eliminate country-specific fixed-effects and then compared to specifications augmented by spatially weighted explanatory variables. Incorporating these variables has a major impact on the perceived reasons why emissions change over time and space. At least for some pollutants it appears that national emissions per capita are strongly influenced by what goes on elsewhere. The implication of this paper is that researchers exploring the EKC hypothesis ought to take greater account of potential spatial relationships.

The remainder of the paper is organised as follows. Section 2 describes spatial econometric techniques. Section 3 describes the data used in the empirical part of the paper. Section 4 presents an econometric model of the EKC augmented by spatially weighted variables and discusses the results obtained. Section 5 concludes with ideas for additional research using spatial econometric techniques.

2. Spatial econometric techniques

Given the existence of several recent reviews of the spatial econometrics literature the review provided here will be brief. For a straightforward introduction to spatial econometrics see [33,22] whilst for more detailed texts see [2,13,16,49]. A collection of interesting papers can be found in Anselin and Florax [7]. Anselin [5] and Florax and van der Vlist [23] provide a recent overview of the literature. For an important recent contribution see [6].

Spatial data may present itself in a variety of forms. Data can be observed at irregularly spaced points on a two-dimensional surface or can alternatively, refer to geographical areas some of which may be contiguous. Data relating to geographic areas can also be analysed as if it related to irregularly spaced points on a two-dimensional plane and this is the approach that I take throughout the remainder of the paper.

Spatial relationships can be modelled in a variety of ways although modelling several forms of spatial relationships at once can lead to problems of identification. A system of spatial lags hypothesises that the value of the dependent variable observed at a particular location is partially determined by a spatially weighted (or “neighbourhood”) average of the value of the dependent variable. Such a model cannot be estimated by ordinary least squares because of the problem of simultaneity bias. It must instead be estimated using either IV estimators or ML techniques. The spatial lag model in matrix form is given by

$$Y = \alpha + \beta X + \rho WY + e,$$

where Y is a vector of dependent variables, X is a matrix of explanatory variables, W is the spatial weight matrix, α and ρ are scalar parameters and β is a vector of parameters. An alternative form of spatial relationship occurs when the dependent variable can be predicted as a function of spatially lagged values of the independent variables. This model can be safely estimated using OLS. The spatial regression model is given by

$$Y = \alpha + \beta X + \gamma WX + e,$$

where γ is a scalar parameter. A third way of incorporating spatial relationships is through spatial dependence in the error term. This refers to a situation in which the errors associated with any one observation are a

⁴Although spatial econometrics techniques have not so far been employed in the estimation of EKC's such techniques are nevertheless becoming more common in the analysis of residential land markets. For a recent attempt to determine the implicit value to households of air quality involving the use of spatial econometric techniques see [31]. According to Florax and van der Vlist [23] spatial econometrics remains an underutilised technique in environmental and natural resource economics.

⁵The paper by Rupasingha et al. [42] uses US county level data to analyse toxic releases. To anticipate some of the terminology used later in the paper, their work considers only spatial error dependence and not the possibility of a spatial lag in either the dependent or the independent variables. All the alternatives are considered in this paper.

spatially weighted average of the errors at nearby sites plus a random error component.⁶ Such models can once more be estimated through maximum likelihood techniques. The spatial error model is given by

$$Y = \alpha + \beta X + e,$$

where

$$e = \lambda We + u$$

and λ is a scalar parameter.

All of these ways of modelling spatial relationships require one to specify in advance the spatial weights matrix. One assumption commonly encountered in empirical work is that the spatial weights matrix should be based on the inverse geographical distances between countries' centroids. It is also sometimes possible to construct a contiguity matrix indicating which countries share a land border with which others. None of these assumptions are without limitations. The use of centroids presents difficulties for large and oddly shaped countries.⁷ And spatial weights matrices based on the existence of land borders would suggest that patently connected countries like Denmark and Sweden are unrelated. Island states moreover have no neighbours and this precludes the use of spatial weights matrices based on the concept of contiguity in this paper since every country must have at least one neighbour.⁸ Countries can also be connected to one another in ways unrelated to distance, in an environmental context, e.g., by virtue of one country being downwind or downstream of another.^{9,10} Weight matrices are commonly row standardised (although this is not essential) such that their rows sum to unity (in order to compute neighbourhood averages) and have zeros along the leading diagonal.

Spatial relationships in the data violate the assumptions underlying OLS leading either to inefficiency and invalid hypothesis testing procedures in the case of spatial error dependence, or to bias and inconsistent parameter estimates in the case of spatial regression and spatial lags.

The specification of spatial models can proceed either on the basis of diagnostic tests or on the basis of theoretical reasoning if strong priors exist. But whilst several tests are available indicating whether spatial relationships are present in data some of these tests have difficulty in discriminating between different forms of spatial relationships. A further problem occurs if the data set to be analysed refers only to a subset of all possible countries since, depending on the spatial weighting scheme adopted, this has implications for the calculation of spatially weighted variables. This problem is particularly relevant to the estimation of EKC's and is discussed below in more detail.

3. Data sources

In a spatial econometric analysis, dealing with missing values is not simply a case of dropping the observation in question. The problem is that, depending on the precise nature of the spatial weight matrix employed, if just one country has a missing value then some or all of the spatially weighted variables cannot be created.¹¹ Since in applied research it is inevitable that data will be unavailable for one or more countries this creates a difficulty especially for weighting schemes based on inverse distance. The recent break up of a number of Eastern European countries compounds this dilemma.

One solution to this problem is simply to fill in missing values and another to limit attention to those weighting systems that are less affected by missing values. The third solution to the problem and the one that I

⁶This model can also be approximated by an autoregressive lag in both the dependent and independent variables.

⁷It is not even guaranteed that a country's centroid will fall within its geographical boundaries.

⁸There exists a database called "The Correlates of War" indicating which countries share common land borders but following the upheavals in Eastern Europe this database is now out of date. Although this database could with a little effort be updated it would not solve the problem of island states.

⁹For Europe anyway scientific evidence is available from the EMEP website in the form of a blame matrix giving that fraction of a country's emissions deposited in any other country. The website address is: http://www.emep.int/SR_data/sr.html. I have investigated [36] whether information on the inclusion of scientific information relating to the atmospheric dispersion of sulphur improves our ability to explain geographical and temporal variations in sulphur emissions in Europe over the period 1951–1990 and find that it does not. This paper uses the concept of contiguity to define the spatial weight matrix.

¹⁰For an interesting discussion on the creation of spatial weights matrices and for links to potentially useful databases see [26].

¹¹Ignoring the problem results in errors-in-variables bias. For more discussion see [16].

Table 1
The data set

Variable	Year	Mean	Std. Dev.	Minimum	Maximum
Per capita SO ₂ emissions (tons)	1990	0.034	0.059	0.000	0.525
	1995	0.026	0.034	0.000	0.211
Per capita NO _x emissions (tons)	1990	0.024	0.024	0.002	0.173
	1995	0.022	0.022	0.002	0.151
Per capita VOC emissions (tons)	1990	0.037	0.038	0.004	0.233
	1995	0.038	0.047	0.004	0.338
Per capita CO emissions (tons)	1990	0.270	0.362	0.009	2.896
	1995	0.253	0.338	0.012	2.530
GDP per capita (1996 international \$)	1990	7287	7070	494	26,892
	1995	7784	7744	321	34,372
HELSINKI		0.118	0.324	0	1
SOFIA		0.162	0.370	0	1

Source: See text.

prefer is to restrict the analysis to those data sets where the data is relatively complete in terms of its geographic coverage. I use estimates of countries' emissions (EMIT) prepared for the World Resources Institute for the years 1990 and 1995.¹² Emissions data are available for sulphur dioxide, nitrogen oxides, non-methane VOCs and carbon monoxide. These emissions are measured in thousands of metric tons. Population estimates (POP) are taken from the same source and are also measured in terms of thousands. These data are then merged with data on GDP per capita measured in constant price 1996 international dollars using the Laspeyres index.¹³ The merged database (see Table 1) contains observations for 135 countries. Apart from Russia virtually all important countries are included and the resulting database is analysed as if it represented a complete list of countries.¹⁴

In this analysis, a variety of spatial weights matrices are constructed from country centroids using great circles distances calculated via the Haversine formula. The coordinates for the centroids of the countries are taken from [3]. Four alternative spatial weights matrices are considered. In the first of these, the spatial weights matrix is calculated using the inverse distance between the centroids of countries as the elements. The matrix is then row standardised. The second spatial weights matrix uses inverse distance weights but is not row standardised. The third spatial weights matrix is based on the concept of the nearest neighbour again judged by the distance between country centroids. The final spatial weights matrix (and the one that ultimately proves to be of most use) defines countries to be neighbours if the distance between their centroids is less than 1750 miles.¹⁵ This matrix is also row standardised.

Two dummy variables (HELSINKI and SOFIA) indicate whether or not a country was a signatory to the 1985 Helsinki protocol limiting sulphur emissions to 30% of their previous totals by 1993. This protocol entered into force in 1987. The other dummy variable records whether the country was a signatory to the 1988 Sofia protocol requiring countries to return their nitrogen oxide emissions to 1987 levels by 1994. It entered into force in 1991. Signatories to these protocols are obviously geographically clustered (see Table 2) and I want to illustrate that any spatial relationships encountered in the data are not due to localised participation in

¹²I am using the EDGAR emissions data. More information on the data can be found in Olivier and Berndowski [39]. The emissions data itself are available at: <http://earthtrends.wri.org/searchable_db/index.cfm?theme=3>.

¹³The GDP data are taken from the Penn World Tables version 6.1 [28].

¹⁴I also considered using the Lefohn et al. [32] data for sulphur emissions but this data set covers only 73 different countries. Another problem with this database is that it finishes in 1990 and particularly for Eastern European countries estimates of purchasing power parity GDP per capita from the Penn World Tables are more generally available only for recent years.

¹⁵The average distance between countries included in the data set is 4837 miles. The maximum distance between countries is 12,368 miles whereas the minimum distance is 40 miles. The most remote country is 1718 miles from its nearest neighbour and it is this country that dictates the choice of 1750 miles as a cut-off point. Under this definition it turns out that countries have on average 19 neighbours. Tunisia has 38 neighbours whereas Canada, Fiji, New Zealand and The Philippines have only one.

Table 2
Tests of spatial autocorrelation

Variable	Moran's I	Geary's C
$\Delta \text{Log}(\text{GDP}/\text{POP})$	0.255 (Prob = 0.000)	0.760 (Prob = 0.000)
$\Delta \text{Log}(\text{SO}_2/\text{POP})$	0.125 (Prob = 0.000)	0.846 (Prob = 0.000)
$\Delta \text{Log}(\text{NO}_x/\text{POP})$	0.250 (Prob = 0.000)	0.725 (Prob = 0.000)
$\Delta \text{Log}(\text{VOC}/\text{POP})$	0.044 (Prob = 0.140)	0.823 (Prob = 0.000)
$\Delta \text{Log}(\text{CO}/\text{POP})$	0.032 (Prob = 0.248)	0.816 (Prob = 0.000)
HELSINKI	0.364 (Prob = 0.000)	0.624 (Prob = 0.000)
SOFIA	0.542 (Prob = 0.000)	0.460 (Prob = 0.000)

Source: See text. Spatial weights matrix = countries within 1750 miles row standardised.

these protocols. The same tests of spatial association demonstrate that the changes in the log of sulphur dioxide and nitrogen dioxide emissions are also geographically clustered, as is the change in the log of GDP per capita.¹⁶

I also employed a localised version of Moran's test [12] which gives an indication of spatial clustering around individual observations. The most interesting thing to emerge from the analysis is that, although neighbouring countries generally share the same trends in emissions, out of all 135 countries the UK presents the leading example of a country achieving modest reductions in sulphur dioxide emissions despite being surrounded by countries achieving a great deal more.¹⁷ Lithuania also falls into this category as does Ecuador which is heavily dependent on oil.

4. Econometric analysis

The majority of EKC's posit a relationship between the logarithm of countries' emissions per capita, the logarithm of purchasing power parity GDP per capita and its squared value [45]. Since the data set includes repeated observations, a time trend (YEAR) is added to account for time-dependent technological progress along with country-specific fixed-effects to obtain

$$\begin{aligned} \text{Log}(\text{EMIT}/\text{POP})_{it} = & \alpha_i + \beta_1 \text{Log}(\text{GDP}/\text{POP})_{it} \\ & + \beta_2 \text{Log}(\text{GDP}/\text{POP})_{it}^2 + \beta_3 \text{YEAR}_t + e_{it}. \end{aligned}$$

I limit attention to examining whether spatial relationships exist for conventionally defined EKC's of the form described above except that, given the existence of only 2 years of data (1990 and 1995), the equation is differenced in order to remove the country-specific intercepts. Dummy variables recording whether the country in question was a signatory to the Helsinki and Sofia protocols are added to the equations describing sulphur dioxide and nitrogen oxides, respectively. The equation for sulphur dioxide emissions actually estimated is, therefore,¹⁸

$$\begin{aligned} \Delta \text{Log}(\text{EMIT}/\text{POP})_i = & \beta_1 \Delta \text{Log}(\text{GDP}/\text{POP})_i + \beta_2 \Delta \text{Log}(\text{GDP}/\text{POP})_i^2 \\ & + \beta_3 \Delta \text{YEAR} + \beta_4 \text{HELSINKI}_i + e_i, \end{aligned}$$

¹⁶For details on Moran's I and Geary's C tests see [2,13,16,49]. Other spatial weight matrices result in a very similar set of findings.

¹⁷In fact, the UK refused to sign the Helsinki protocol and was called the "dirty man of Europe" as a consequence, so this result comes as no real surprise.

¹⁸The equations for the remaining pollutants are analogous apart from the SOFIA dummy for nitrogen oxides and the absence of in force protocols for non-methane VOCs and carbon monoxide.

where Δ indicates that a variable has been differenced. The equation is estimated by OLS for each pollutant in turn and the results are presented in Table 3.¹⁹

For sulphur dioxide there appears to be a monotonic relationship between GDP per capita and emissions per capita. Removing the statistically insignificant higher-order term for income per capita reveals a statistically significant income elasticity of 0.47. According to this model any reductions in sulphur emissions have been the result of time-dependent technological progress. The results for nitrogen oxides also indicate that time-dependent technological progress plays an important role and that, rather than taking an inverted U shape, emissions per capita increase with respect to GDP per capita over virtually the entire range of per capita incomes (the turning point is \$410 per capita).

The theoretical work of Barrett suggests that environmental agreements involving large numbers of participants are unlikely to improve on the non-cooperative outcome [10]. Nonetheless, participation in the Helsinki protocol appears to decrease emissions per capita emissions of sulphur dioxide although the effect is significant only at the 10% level of significance. The Sofia protocol by contrast appears to have had a highly significant impact on the per capita emissions of nitrogen oxides, even if the two protocols' effects were similar in magnitude.²⁰

For non-methane VOCs and carbon monoxide it appears that emissions are likewise increasing with respect to income levels over virtually the whole range of per capita income, and that any improvement in emissions per capita comes from time-dependent changes in technology. The turning point for VOCs is \$627 and for carbon monoxide \$785 per capita. Although it is the first time that this particular data set has been used to estimate and EKC these findings are not inconsistent with the accumulated evidence. In a recent literature review, Stern [45] states that: "The only robust conclusions from the EKC literature appear to be that...emissions tend to be monotonic in income." (p. 1426).

I now perform a variety of tests in order to determine whether spatial relationships are present in the regressions and if so what form they take. The Robust LM test for spatial error dependence checks for the existence of a spatial error dependence even in the presence of a spatial lag. The Robust LM test for error dependency, however, is adversely affected by the presence of non-normality whose presence is detected by the Jarque–Bera test. In such circumstance, the Kelejian–Robinson [29] test provides a consistent alternative albeit one which requires large samples before it can be used reliably. I present both test statistics as well as Robust LM tests for spatial lag dependence. For technical details on all these tests see [4].

The Robust LM tests for a spatial lag are significant for both sulphur dioxide and nitrogen oxide emissions for three out of four spatial weight matrices. The test for a spatial lag is statistically significant for carbon monoxide only for the spatial weights matrix based on inverse distances. The Kelejian and Robinson test suggests that spatial error dependence is present only in the case of sulphur dioxide and then only when the spatial weights matrix is based on the nearest neighbour concept.

Given this set of results the equations for sulphur dioxide and nitrogen oxides were reanalysed allowing for the existence of a spatial lag calculated using the row standardised weight matrix with the cut-off point of 1750 miles (see Table 4). Although the use of other spatial weights matrices also leads to the conclusion that there exists a highly significant spatial lag for these pollutants, this particular spatial weights matrix possesses three important advantages. Firstly, the coefficients that emerge from the subsequent regression can be readily interpreted by virtue of being row standardised. Secondly, a problem that sometimes afflicts IV estimators of spatial lag models, namely that estimates of the spatial lag coefficient are not bound to take a value within the unit interval, does not occur with this spatial weights matrix. When the estimated value of the spatial lag parameter falls outside this interval others have interpreted this as evidence of model misspecification. Thirdly, the fact that this spatial weights matrix is based on absolute rather than relative distances suggests that it is best placed to capture distance related phenomena.

¹⁹These calculations are performed using SPACESTAT version 1.90.

²⁰Following a referee's suggestion I tried including the dummy variable for the Sofia protocol in the equation for sulphur dioxide and the dummy variable for the Helsinki protocol in the nitrogen oxides equation. Including the Helsinki dummy in the equation for nitrogen oxides results in a statistically significant coefficient. To the extent that compliance with these protocols requires different technological approaches (scrubbing for sulphur dioxide versus burner adjustment for nitrogen oxides) one would not expect a country's agreeing to reduce its sulphur dioxide emissions to also reduce its nitrogen oxide emissions. This indicates that being a 'protocol-loving' country is associated with achieving greater reductions in both pollutants, not just the pollutant that is the subject of the protocol.

Table 3
The ordinary least-squares regression model

	Sulphur dioxide	Sulphur dioxide	Nitrogen oxides	Non-methane VOCs	Carbon monoxide
$\Delta\text{Log}(\text{GDP}/\text{POP})$	−0.604 (−0.547)	0.478 (3.400)	−1.480 (−2.397)	−0.992 (−2.108)	−1.560 (−2.302)
$\Delta\text{Log}(\text{GDP}/\text{POP})^2$	0.067 (0.988)		0.123 (3.263)	0.077 (2.685)	0.117 (2.813)
ΔYear	−0.022 (−3.561)	−0.020 (−3.427)	−0.008 (−2.439)	−0.005 (−2.268)	−0.017 (−4.981)
HELSINKI	−0.156 (−1.836)	−0.163 (−1.928)			
SOFIA			−0.182 (−4.417)		
<i>F</i> -test	6.054 (Prob = 0.000)	8.594 (Prob = 0.000)	26.249 (Prob = 0.000)	13.206 (Prob = 0.000)	11.382 (Prob = 0.000)
Jarque–Bera	1437.221 (Prob = 0.000)	1435.333 (Prob = 0.000)	72.991 (Prob = 0.000)	1224.680 (Prob = 0.000)	3433.153 (Prob = 0.000)
White	1.768 (Prob = 0.987)	0.538 (Prob = 0.969)	16.578 (Prob = 0.034)	0.494 (Prob = 0.992)	0.370 (Prob = 0.996)
Spatial weights matrix = inverse distance row standardised					
Robust LM error	3.267 (Prob = 0.070)	3.341 (Prob = 0.067)	0.895 (Prob = 0.343)	0.534 (Prob = 0.464)	1.432 (Prob = 0.231)
Kelejian–Robinson	2.531 (Prob = 0.639)	2.574 (Prob = 0.461)	5.423 (Prob = 0.246)	1.361 (Prob = 0.714)	1.142 (Prob = 0.766)
Robust LM lag	5.435 (Prob = 0.019)	5.589 (Prob = 0.018)	6.952 (Prob = 0.008)	0.365 (Prob = 0.545)	1.471 (Prob = 0.225)
Spatial weights matrix = inverse distance not row standardised					
Robust LM error	2.149 (Prob = 0.142)	2.203 (Prob = 0.137)	0.457 (Prob = 0.498)	1.304 (Prob = 0.253)	5.297 (Prob = 0.021)
Kelejian–Robinson	2.531 (Prob = 0.639)	2.574 (Prob = 0.461)	5.423 (Prob = 0.246)	5.736 (Prob = 0.219)	1.142 (Prob = 0.766)
Robust LM lag	6.472 (Prob = 0.010)	6.607 (Prob = 0.010)	10.944 (Prob = 0.000)	0.802 (Prob = 0.370)	7.452 (Prob = 0.006)
Spatial weights matrix = nearest neighbour					
Robust LM error	0.453 (Prob = 0.500)	0.849 (Prob = 0.356)	0.163 (Prob = 0.686)	1.137 (Prob = 0.286)	1.004 (Prob = 0.316)
Kelejian–Robinson	13.873 (Prob = 0.007)	15.723 (Prob = 0.001)	6.322 (Prob = 0.176)	4.603 (Prob = 0.330)	0.063 (Prob = 0.995)
Robust LM lag	1.160 (Prob = 0.281)	1.791 (Prob = 0.180)	1.378 (Prob = 0.240)	0.858 (Prob = 0.354)	1.178 (Prob = 0.277)
Spatial weights matrix = countries within 1750 miles row standardised					
Robust LM error	3.531 (Prob = 0.060)	4.026 (Prob = 0.044)	1.759 (Prob = 0.184)	1.824 (Prob = 0.176)	1.045 (Prob = 0.306)
Kelejian–Robinson	3.010 (Prob = 0.556)	3.361 (Prob = 0.339)	1.958 (Prob = 0.743)	5.709 (Prob = 0.126)	0.719 (Prob = 0.868)
Robust LM lag	5.594 (Prob = 0.018)	6.111 (Prob = 0.013)	5.740 (Prob = 0.016)	1.674 (Prob = 0.195)	0.976 (Prob = 0.323)
R^2	0.121	0.115	0.375	0.154	0.147
Observations	135	135	135	135	135

Note: Unless otherwise stated the figures in parentheses are *t*-statistics.

Dependent variable = $\Delta\text{Log}(\text{EMIT}/\text{POP})$.

Method = ordinary least squares.

Table 4
The model with a spatial lag

	Sulphur dioxide	Nitrogen oxides
$\Delta\text{Log}(\text{GDP}/\text{POP})$	0.264 (1.523)	−1.597 (−2.630)
$\Delta\text{Log}(\text{GDP}/\text{POP})^2$		0.126 (3.012)
ΔYear	−0.002 (−0.844)	−0.007 (−2.227)
HELSINKI	−0.002 (−0.018)	
SOFIA		−0.134 (−2.200)
$\Sigma w_i \Delta\text{Log}(\text{EMIT}/\text{POP})$	0.848 (2.659)	0.466 (2.844)
R^2	0.176	0.470
Observations	135	135

Note: Unless otherwise stated the figures in parentheses are *t*-statistics.

Dependent variable = $\Delta\text{Log}(\text{EMIT}/\text{POP})$.

Method = robust instrumental variables.

Spatial weights matrix = countries within 1750 miles row standardised.

In view of the twin problems of non-normality and heteroscedasticity, the regression equations for each of these two pollutants are estimated using robust IV techniques. It appears that a 1% change in neighbourhood average per capita sulphur dioxide emissions results in a 0.84% change in per capita emissions. For nitrogen oxides the corresponding figure is 0.46. These results reveal the surprising extent to which changes in national emissions per capita are influenced by changes in emissions per capita in neighbouring countries.²¹ It furthermore appears that, at least in the case of the equation for sulphur dioxide, the inclusion of a spatial lag has a devastating effect on the statistical significance of the remaining parameters none of which are now statistically significant.^{22,23} By contrast the parameters of the equation for nitrogen oxides are similar to those displayed in Table 3 despite the presence of a statistically significant spatial lag. To gain a further sense of the importance of spatial effects, note that the elasticity of per capita emissions of sulphur with respect to neighbourhood per capita emissions greatly exceeds the income elasticity of per capita emissions.

I also estimated by OLS a version of the model in which the equations were augmented by a spatially weighted GDP per capita where the neighbourhood is once more defined as all other countries within a 1750 mile radius. The purpose of these regressions is to test the hypothesis that being a high-income country relative to ones neighbours decreases per capita emissions. The coefficient on this variable should be positive for the hypothesis to be upheld. This variable is, however, uniformly negative for all four pollutants although its inclusion did not improve significantly the fit of any apart from nitrogen oxides, where the effect is quite marked (see Table 5).²⁴ More specifically a 1% change in neighbourhood income causes a 0.43% change in per capita emissions. It is also interesting that when this term is added to the equation describing per capita emissions of nitrogen oxides that the evidence favouring the existence of a spatial lag disappears. For sulphur

²¹This refers only to the direct impact. Changes in national emissions affect neighbourhood average emissions setting up a feedback effect. For details on how to calculate the full impact multiplier for a unit change in the dependent variable see [31].

²²These results are not inconsistent with those of Stern [44], who finds that the sulphur and GDP per capita space-time surfaces for Europe do not appear to be related to one another.

²³I would instead attribute the observed reductions in sulphur dioxide emissions in Europe to a politically sanctioned toughening of environmental regulations and the installation of end of the pipe abatement technologies in the Netherlands and Germany spilling over into nearby countries [19].

²⁴Using the other spatial weight matrices to calculate the spatially lagged values for income per capita did not result in statistically significant coefficients.

Table 5
Spatial regressions

	Sulphur dioxide	Nitrogen oxides (Robust standard errors)	Non-methane VOCs	Carbon monoxide
$\Delta\text{Log}(\text{GDP}/\text{POP})$	0.374 (2.385)	−1.710 (−2.595)	−1.074 (−2.269)	−1.592 (−2.320)
$\Delta\text{Log}(\text{GDP}/\text{POP})^2$		0.131 (2.891)	0.080 (2.776)	0.118 (2.821)
ΔYear	−0.021 (−3.606)	−0.010 (−3.374)	−0.006 (−2.440)	−0.018 (−4.962)
HELSINKI	−0.158 (−1.873)			
SOFIA		−0.171 (−2.856)		
$\Delta\Sigma w_i\text{Log}(\text{GDP}/\text{POP})$	−0.409 (−1.449)	−0.438 (−3.071)	−0.158 (−1.312)	−0.061 (−0.348)
<i>F</i> -test	6.478 (Prob = 0.000)	22.704 (Prob = 0.000)	9.426 (Prob = 0.000)	7.577 (Prob = 0.000)
Jarque–Bera	1164.418 (Prob = 0.000)	66.527 (Prob = 0.000)	1226.938 (Prob = 0.000)	3388.548 (Prob = 0.000)
White	10.955 (Prob = 0.204)	46.772 (Prob = 0.000)	0.516 (Prob = 0.915)	3.056 (Prob = 0.962)
Robust LM error	4.117 (Prob = 0.042)	0.000 (Prob = 0.981)	0.003 (Prob = 0.953)	1.718 (Prob = 0.189)
Kelejian–Robinson	7.464 (Prob = 0.113)	2.511 (Prob = 0.774)	6.581 (Prob = 0.159)	0.972 (Prob = 0.913)
Robust LM lag	4.770 (Prob = 0.028)	0.000 (Prob = 0.978)	0.033 (Prob = 0.855)	1.651 (Prob = 0.198)
R^2	0.129	0.411	0.177	0.128
Observations	135	135	135	135

Note: Unless otherwise stated the figures in parentheses are *t*-statistics.

Dependent variable = $\Delta\text{Log}(\text{EMIT}/\text{POP})$.

Method = ordinary least squares.

Spatial weights matrix = countries within 1750 miles row standardised.

dioxide on the other hand, the test for the existence of a spatial lag is still significant. It might be that the spatial regression model is more suitable for nitrogen oxides than the spatial lag model.

Last of all the results presented in Table 3 suggested that the choice between the spatial lag and spatial error model might depend on the assumptions made regarding the nature of the spatial weights matrix.²⁵ In particular, using a spatial weights matrix based on the nearest neighbour concept there appeared to be evidence of spatial error dependence for changes in sulphur dioxide emissions. Estimating a spatial error model using ML methods reveals that autocorrelation is significantly different from zero but with a *t*-statistic of only 1.967 this result is not very compelling (see Table 6).

5. Conclusions

Spatial econometric techniques provide researchers with a means to explore the question of whether or not the environmental performance of countries measured in terms of emissions per capita is dependent on the characteristics of neighbouring states. I find that per capita emissions for two important pollutants are a function of the spatially weighted average of the emissions per capita of neighbouring countries. Since

²⁵This occurrence is not unusual [2].

Table 6
The model with a spatial error

	Sulphur dioxide
$\Delta\text{Log}(\text{GDP}/\text{POP})$	0.414 (2.834)
ΔYear	−0.021 (−3.227)
HELSINKI	−0.122 (−1.370)
λ	0.133 (1.967)
R^2	0.080
Observations	135

Note: Unless otherwise stated the figures in parentheses are *t*-statistics.

Dependent variable = $\Delta\text{Log}(\text{EMIT}/\text{POP})$.

Method = maximum likelihood.

Spatial weights matrix = nearest neighbour.

conventional estimation techniques yield biased and inconsistent parameter estimates when a spatial lag is incorrectly omitted such findings also provide a possible explanation of why estimating EKC's on geographically defined subsets of data results in evidence of parameter instability.

There is no support here for the suggestion that environmental problems are resolved automatically by economic growth [11]. There is instead a mechanism by which changes in emissions are transmitted to neighbouring countries. But it is unclear whether this is due to governments mimicking each other environmental policy or whether these are purely incidental effects caused by the geographical diffusion of technologies, products and lifestyles. But it may be significant that spatial effects are most apparent for precisely those pollutants which have been the focus of the majority of regulatory attention.

Concern has frequently been expressed that what the existing evidence reveals might not be that economic growth results in an improvement in environmental quality, but rather the international displacement of polluting activities to countries with lower environmental standards and lower per capita incomes. I find evidence to suggest that having high-income neighbours is associated with having much lower per capita nitrogen oxide emissions.

On the basis of these results it appears that researchers who do not account for spatial relationships risk misspecification and are in danger of presenting a largely incorrect story of the reasons why countries' emissions have changed over time.

A large number of research questions now present themselves. There are many other pollutants waiting to be explored using spatial econometric techniques, some of which are purely local in nature whilst others are perfectly mixing. Do changes in the per capita emissions of these pollutants also generate international spillovers? In instances where it can be applied, does geographical contiguity provide a better representation of spatial relationships between countries than distance-based measures? To what extent do unacknowledged spatial relationships account for the parameter instability reported in existing studies? Do national reductions in carbon emissions lead to compensating increases in carbon emissions elsewhere? Given time lags in decision-making future analyses should also test for the existence of a spatio-temporal lag rather than the simultaneous spatial lag presented in this paper.

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